

- Q.28 What is the texturing process, and what are its different types?
- Q.29 Discuss the importance of moisture control in man-made fiber processing and how this influences machinery settings.
- Q.30 What is Drawing? Why it is done in two stages?
- Q.31 What are the important fibre characteristics required for blending?
- Q.32 Write objective of Heat Setting.
- Q.33 Explain the information and objectives of post-spinning operations in man-made fiber production.
- Q.34 Explain Dyeing of cotton with reactive dye
- Q.35 Discuss the role of segregation in textile production. How does it affect the final quality of the product?

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Explain the Air-Jet Spinning process in detail, including the mechanisms involved in fiber separation, twisting, and winding of the yarn. Discuss how the air-jet system controls fiber orientation and yarn structure.
- Q.37 Compare at least 10 and contrast the yarn properties of Ring, Rotor, Air-jet, and Friction Spinning.
- Q.38 Explain the process, equipment used, and the key objectives of heat setting. How does heat setting improve fiber performance and longevity?

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Roll No.

6th Sem / Textile Technology
Subject:- Advanced Yarn Manufacturing/
Modern Methods in Yarn Product

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Ring spinning is not suitable for:
 a) Very fine yarns b) Very coarse yarns
 c) Medium count yarns d) Blended yarns
- Q.2 The main limitation of ring spinning is:
 a) Low yarn quality
 b) High spindle and traveller friction
 c) High production rate
 d) Ability to spin coarse yarns
- Q.3 Fibre fineness is usually measured in:
 a) Tex b) Denier
 c) Micronaire value d) All of the above
- Q.4 The heart of the rotor spinning machine is:
 a) Drafting zone b) Rotor
 c) Navel d) Opening roller
- Q.5 Air-jet spinning uses which medium for yarn formation?
 a) Water jet b) Frictional discs
 c) Air vortex d) Ring and traveler

- Q.6 Which type of yarn is generally bulky and low strength?
 a) Ring b) Rotor
 c) Air-jet d) Friction
- Q.7 Heat setting imparts:
 a) Permanent dimensional stability
 b) High hairiness
 c) Random molecular orientation
 d) Lower strength always
- Q.8 Which spinning system gives the strongest yarns?
 a) Rotor b) Air-jet
 c) Ring d) Friction
- Q.9 Which spinning system is most suitable for man-made fibre blends?
 a) Ring spinning b) Rotor spinning
 c) Friction spinning d) All of the above
- Q.10 For good blending, fibres should have:
 a) Equal dyeing behaviour
 b) Unequal tensile strength
 c) High hairiness
 d) Different crimp levels only

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Why is fibre fineness important in spinning?
 Q.12 What is the main limitation of ring spinning related to spindle speed?

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- Q.13 Ring spinning has _____ productivity compared to rotor or air-jet spinning.
 Q.14 What is the main principle of yarn formation in rotor spinning/
 Q.15 The feed material for rotor spinning is roving. True/False
 Q.16 The twist in air-jet spinning is mainly _____.
 Q.17 The productivity of friction spinning is _____ than rotor spinning.
 Q.18 Name one objective of post spinning operations.
 Q.19 What is false twist?
 Q.20 What is segregation in textiles?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Write disadvantage of ring spinning.
 Q.22 Explain fiber length, strength, cohesion and fiber fineness in air jet yarn.
 Q.23 What are the basic elements are principles of a rotor spinning machine?
 Q.24 Describe the passage of fibers through a rotor spinning frame.
 Q.25 Discuss the range of speeds for the opening roller and rotor in rotor spinning.
 Q.26 Explain advantages of Air-jet Spinning.
 Q.27 Explain the concept of Friction Spinning.

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